

Lesson 9. Partial Differential Equations II

University of Budapest

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The general **second-order linear PDE** has the following form:

$$a(x, y)u_{xx} + b(x, y)u_{xy} + c(x, y)u_{yy} + d(x, y)u_x + e(x, y)u_y + f(x, y)u = g(x, y), \quad (1)$$

where the coefficients a, b, c, d, e, f and the free term g are, in general, functions of the independent variables x, y , but do not depend on the unknown function u .

The Diffusion Equation

The one-dimensional diffusion equation is the linear second-order partial differential equation

$$u_t - Du_{xx} = f(x, t),$$

where $u = u(x, t)$ describes how some quantity changes along x over time t , x is a real space variable, t is a time variable, and D is a positive constant, called the diffusion coefficient.

When $f \equiv 0$, the equation is said to be **homogeneous**:

$$u_t - Du_{xx} = 0.$$

A common example of diffusion is given by **heat conduction** in a solid body.

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The Wave Equation

The one-dimensional wave equation is the linear second-order partial differential equation

$$u_{tt} - c^2 u_{xx} = f(x, t),$$

where $u(x, t)$ is the displacement of a point on the string, t is a time variable, and the coefficient c has the dimensions of speed and represents the wave propagation speed along the string.

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Laplace's or potential equation

The **two**-dimensional Laplace's equation is the linear second-order homogeneous partial differential equation:

$$\Delta u = 0,$$

where $u(x, y)$ is a **scalar field** (e.g., temperature or electric potential) defined over a spatial domain, and Δ is the **Laplacian**, given by

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}.$$

The diffusion and wave equations model evolution phenomena over time, whereas the Laplace equation describes the corresponding **steady state**, in which the solution does not depend on time anymore.

If $f(x, y)$ is a given source function that depends on the spatial variables x and y then the Laplace's equation become

$$\Delta u = f,$$

Poisson's equation, where $u(x, y)$ represents a scalar field, such as electric potential or temperature.

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Classification of PDE's

Given a general second order linear partial differential equation, how can we tell what type it is? This is known as the **classification of second order PDEs**.

We will classify second-order PDEs into three different categories:

- hyperbolic;
- parabolic;
- elliptic.

What do the names hyperbolic, elliptic and parabolic mean?

Consider the quadratic curve

$$ax^2 + 2bxy + cy^2 + fx + gy + h = 0.$$

It represents a hyperbola, an ellipse, or a parabola depending on whether the discriminant, $ac - b^2$, is less than zero, greater than zero, or equal to zero.

These are the conditions for the matrix

$$\begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

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Classification of PDE's

By analogy, the equation

$$a(x, y) \frac{\partial^2 u}{\partial x^2} + 2b(x, y) \frac{\partial^2 u}{\partial x \partial y} + c(x, y) \frac{\partial^2 u}{\partial y^2} + (\text{lower orders}) = 0, \quad (2)$$

is said to be hyperbolic, elliptic, or parabolic at a point (x, y) if

$$\begin{vmatrix} a(x, y) & b(x, y) \\ b(x, y) & c(x, y) \end{vmatrix} = (ac - b^2) \Big|_{(x, y)}$$

is less than, greater than, or equal to zero, respectively.

Classification of PDE's

The Heat Equation

For the heat equation:

$$u_t - Du_{xx} = 0,$$

the corresponding determinant of the coefficient matrix is given by

$$\begin{vmatrix} -D & 0 \\ 0 & 0 \end{vmatrix} = 0.$$

Thus, the **Heat Equation** is **Parabolic**.

Classification of PDE's

The Wave Equation

For the wave equation:

$$u_{tt} - c^2 u_{xx} = 0,$$

the corresponding determinant of the coefficient matrix is given by

$$\begin{vmatrix} 1 & 0 \\ 0 & -c^2 \end{vmatrix} = -c^2 < 0.$$

Thus, the **Wave Equation** is **Hyperbolic**.

Classification of PDE's

The Laplace's Equation

For the Laplace's equation:

$$u_{xx} + u_{yy} = 0,$$

the corresponding determinant of the coefficient matrix is given by

$$\begin{vmatrix} 1 & 0 \\ 0 & 1 \end{vmatrix} = 1 > 0.$$

Thus, the **Laplace's Equation** is **Elliptic**.

Classification of PDE's

Determine the regions in the xy -plane where the following equation is hyperbolic, parabolic, or elliptic.

The equation is given by

$$u_{xx} + yu_{yy} + \frac{1}{2}u_y = 0.$$

The coefficients of the leading terms in this equation are

$$a = 1, \quad b = 0, \quad c = y.$$

The discriminant is then

$$\Delta = b^2 - 4ac = -4y.$$

Hence, the equation is:

- hyperbolic when $y < 0$,
- parabolic when $y = 0$,
- elliptic when $y > 0$.

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Since the equations that we study come from physical considerations, the addition conditions must be physically relevant as well. These conditions generally come in two types: **initial conditions** and **boundary value conditions**.

Initial Conditions

Initial conditions specify the physical state of a system at a specific point in time, t_0 . These conditions are functions of spatial variables that describe the initial station of the system.

Boundary Conditions

Boundary conditions specify the behavior of the solution at the boundaries of the domain. In physical problems, the domain is often restricted to a region where the solutions must live. The boundary of this domain represents where the system interacts with external factors, such as the environment, or external forces, and boundary conditions are needed to account for these interactions.

In higher dimensions, the domain can be bounded by curves (2D), surfaces (3D), or more complex geometric shapes.

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In higher dimensions, the domain can be bounded by curves (2D), surfaces (3D), or more complex geometric shapes.

There are three primary types of boundary conditions:

- **Dirichlet Boundary Conditions (D):** The value of the solution $u(x, t)$ is specified on the boundary.

$$u(x, t)|_{\partial D} = f(x, t).$$

- **Neumann Boundary Conditions (N):** The normal derivative (rate of change) of the solution is specified on the boundary.

$$\frac{\partial u}{\partial n}|_{\partial D} = g(x, t),$$

where $\frac{\partial u}{\partial n}$ denotes the derivative normal to the boundary ∂D .

- **Robin Boundary Conditions (R):** A linear combination of the solution and its normal derivative is specified on the boundary.

$$\frac{\partial u}{\partial n} + a(x, t)u(x, t)|_{\partial D} = h(x, t),$$

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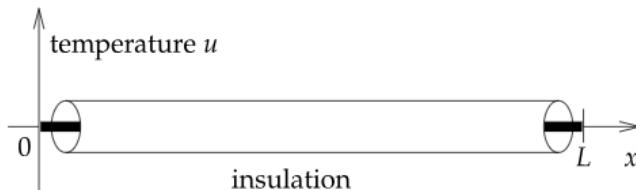
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Insulated wire

Consider a wire (or a thin metal rod) of length with insulated lateral surface (except at the endpoints) and no internal sources of heat. Let x denote the position along the wire and let t denote time.



Temperature at point x at time t

To describe the problem, let $u(x, t)$ ($0 < x < L$, $t > 0$) represent the temperature of the point x of the wire at time t .

Initial Condition

Given that the initial temperature distribution of the wire is

$$u(x, 0) = f(x) \quad \text{for } 0 < x < L$$

for some known function $f(x)$.

Boundary Conditions

Given that the ends of the wire are held at constant temperature 0

$$u(0, t) = 0 \quad \text{and} \quad u(L, t) = 0 \quad \text{for all } t > 0$$

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$$u(0, t) = 0 \quad \text{and} \quad u(L, t) = 0 \quad \text{for all } t > 0$$

What is $u(x, t)$ for $0 < x < L, t > 0$?

To answer on this question we said that u satisfies the one-dimensional heat equation:

$$u_t = ku_{xx}$$

where k is a constant (the thermal conductivity of the material).

The **initial boundary value problem with homogeneous Dirichlet boundary conditions** for the one-dimensional heat equation is given by:

$$u_t = ku_{xx}, \quad 0 < x < L, t > 0$$

$$u(0, t) = u(L, t) = 0, \quad t > 0$$

$$u(x, 0) = f(x), \quad 0 < x < L.$$

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$$u(x, 0) = f(x), \quad 0 < x < L.$$

Separation of variables

The **method of separation of variables** is to try to find solutions that are products of functions of one variable:

$$u(x, t) = X(x)T(t).$$

Plug $u(x, t) = X(x)T(t)$ into the heat equation to obtain

$$X(x)T'(t) = kX''(x)T(t)$$

or

$$\frac{T'(t)}{kT(t)} = \frac{X''(x)}{X(x)}.$$

Notice that the left-hand side does not depend on x and the right-hand side does not depend on t . Thus, each side must be a constant. Let us call this constant $-\lambda^2$.

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Separation of variables

We obtain the two equations

$$\frac{T'(t)}{kT(t)} = -\lambda^2 = \frac{X''(x)}{X(x)}.$$

In other words,

$$X''(x) + \lambda^2 X(x) = 0,$$

$$T'(t) + \lambda^2 kT(t) = 0.$$

We are looking for nontrivial solutions X of the problem

$$X'' + \lambda^2 X = 0, X(0) = 0, \quad X(L) = 0.$$

To solve the eigenvalue problem with the differential equation and boundary conditions given, let's analyze the three cases for λ^2 and derive solutions accordingly.

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Separation of variables

Case 1: $\lambda^2 < 0$

If $\lambda^2 < 0$, the differential equation becomes

$$X'' - \lambda^2 X = 0$$

The general solution of this equation is

$$X(x) = C_1 e^{\lambda x} + C_2 e^{-\lambda x}$$

Applying the boundary conditions:

- $X(0) = 0$ implies $C_1 + C_2 = 0$, so $C_1 = -C_2$.
- $X(L) = 0$ gives $C_1 e^{\lambda L} + C_2 e^{-\lambda L} = 0$, which implies $C_1 = C_2 = 0$ for any nontrivial solution.

Thus, for $\lambda^2 < 0$, the only solution is the **trivial solution**: $X(x) = 0$.

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Separation of variables

Case 2: $\lambda^2 = 0$

If $\lambda^2 = 0$, then the equation reduces to

$$X'' = 0$$

The general solution is

$$X(x) = C_1x + C_2$$

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- $X(L) = 0$ implies $C_1L = 0$, so $C_1 = 0$.

Thus, for $\lambda^2 = 0$, the only solution is again the **trivial solution**: $X(x) = 0$.

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Thus, for $\lambda^2 = 0$, the only solution is again the **trivial solution**: $X(x) = 0$.

Separation of variables

Case 3: $\lambda^2 > 0$

If $\lambda^2 > 0$, the differential equation becomes

$$X'' + \lambda^2 X = 0$$

The general solution is

$$X(x) = C_1 \cos(\lambda x) + C_2 \sin(\lambda x)$$

Applying the boundary conditions:

- $X(0) = 0$ implies $C_1 = 0$.
- $X(L) = 0$ gives $C_2 \sin(\lambda L) = 0$.

For a nontrivial solution let's take $C_2 = 1$, then $\sin(\lambda L) = 0$, which requires $\lambda L = n\pi$ for integer n . Thus, $\lambda = \frac{n\pi}{L} \Rightarrow \lambda = \frac{n\pi}{L}$.

The corresponding solutions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

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For a nontrivial solution let's take $C_2 = 1$, then $\sin(\lambda L) = 0$, which requires $\lambda L = n\pi$ for integer n . Thus, $\lambda = \frac{n\pi}{L} \Rightarrow \lambda = \frac{n\pi}{L}$.

The corresponding solutions are

$$X_n(x) = \sin\left(\frac{n\pi x}{L}\right)$$

Separation of variables

Case 3: $\lambda^2 > 0$

If $\lambda^2 > 0$, the differential equation becomes

$$X'' + \lambda^2 X = 0$$

The general solution is

$$X(x) = C_1 \cos(\lambda x) + C_2 \sin(\lambda x)$$

Applying the boundary conditions:

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Separation of variables

The corresponding T_n must satisfy the equation

$$T'_n(t) + \frac{n^2\pi^2}{L^2}kT_n(t) = 0.$$

This is first-order Separable ODE and the solution is just an exponential:

$$T_n(t) = b_n e^{\frac{-n^2\pi^2}{L^2}kt}.$$

Normal mode

Then the product solution :

$$u_n(x, t) = X_n(x)T_n(t) = \sin\left(\frac{n\pi}{L}x\right) b_n e^{\frac{-n^2\pi^2}{L^2}kt}.$$

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Finally, we use superposition to write the solution as

$$u(x, t) = \sum_{n=1}^{\infty} u_n(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right) e^{\frac{-n^2\pi^2}{L^2}kt}.$$

Our next step is to determine b_n using IC:

We note that $u_n(x, 0) = \sin\left(\frac{n\pi}{L}x\right)$. Thus,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi}{L}x\right).$$

Recognizing this sum as half-range sine expansion f we get

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi x}{L} dx$$

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SOLUTION OF THE ONE DIMENSIONAL HEAT EQUATION

The solution of the one dimensional heat boundary value problem

$$\frac{\partial u}{\partial t} = k \frac{\partial^2 u}{\partial x^2} \quad 0 < x < L, \quad t > 0, \quad (1)$$

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is

$$u(x, t) = \sum_{n=1}^{\infty} b_n e^{-\lambda_n^2 k t} \sin \frac{n\pi}{L} x, \quad (4)$$

where

$$b_n = \frac{2}{L} \int_0^L f(x) \sin \frac{n\pi}{L} x \, dx \quad \text{and} \quad \lambda_n = \frac{n\pi}{L}, \quad n = 1, 2, \dots \quad (5)$$

Example

Example:

Consider an insulated wire of length 1 whose ends are embedded in ice (temperature 0). Let $k = 0.003$ and let the initial heat distribution be $u(x, 0) = 50x(1 - x)$. Find the temperature function $u(x, t)$.

Solution:

We are solving the following IBVP problem:

$$\begin{aligned}u_t &= 0.003 u_{xx}, \\u(0, t) &= u(1, t) = 0, \\u(x, 0) &= 50x(1 - x) \quad \text{for } 0 < x < 1.\end{aligned}$$

We find b_n according to the formula (5):

$$b_n = 2 \int_0^1 50x(1 - x) \sin(n\pi x) dx = \frac{200}{\pi^3 n^3} - \frac{200(-1)^n}{\pi^3 n^3} = \begin{cases} 0 & \text{if } n \text{ is even,} \\ \frac{400}{\pi^3 n^3} & \text{if } n \text{ is odd.} \end{cases}$$

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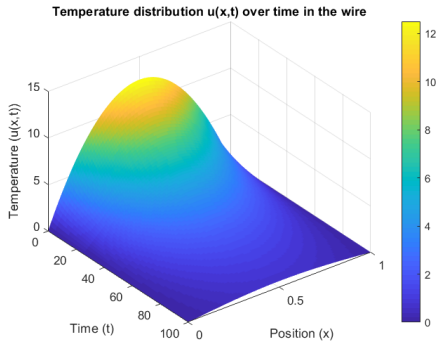
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The solution is then given by the series:

$$u(x,t) = \sum_{\substack{n=1 \\ n \text{ odd}}}^{\infty} \frac{400}{\pi^3 n^3} \sin(n\pi x) e^{-n^2 \pi^2 0.003 t}.$$



Insulated ends

Now suppose the ends of the wire are insulated. In this case the problem of heat transfer in a wire of length L with initial distribution $f(x)$ has the following boundary conditions:

$$u_x(0, t) = 0, \quad u_x(L, t) = 0.$$

The **initial boundary value problem with homogeneous Neumann boundary conditions** for the one-dimensional heat equation is given by:

$$u_t = ku_{xx}, \quad 0 < x < L, t > 0$$

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Separation of variables

Again we try a solution of the form

$$u(x, t) = X(x)T(t).$$

By the same procedure as before we plug into the heat equation and arrive at the following two equations

$$\begin{aligned}X''(x) + \lambda^2 X(x) &= 0, \\T'(t) + \lambda^2 k T(t) &= 0.\end{aligned}$$

Notice that the differential equations are the same as in previous problem, but the boundary conditions are different. This difference in the boundary conditions will give rise to different solutions in X .

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Separation of variables

Case 1: $\lambda^2 < 0$

It is easy to check that negative values of λ^2 lead only to the **trivial solution** $X = 0$.

Case 2: $\lambda^2 = 0$

For $\lambda^2 = 0$, we have $X'' = 0$, leading to $X = c_1x + c_2$. Differentiating and imposing the boundary conditions, we find $c_1 = 0$ and c_2 is arbitrary. Thus $\lambda^2 = 0$ leads to a **nontrivial solution**

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Separation of variables

Case 3: $\lambda^2 > 0$

To consider the case of positive separation constant, we solve the resulting equation, and get $X = c_1 \cos \lambda x + c_2 \sin \lambda x$. Computing

$$X' = -c_1 \lambda \sin \lambda x + c_2 \lambda \cos \lambda x$$

and using the first boundary condition $X'(0) = 0$, we obtain $c_2 = 0$. Using the second boundary condition $X'(L) = 0$, we obtain the equation $\sin \lambda L = 0$, from which we get

$$\lambda = \lambda_n = \frac{n\pi}{L}, \quad n = 1, 2, \dots$$

and hence

$$X_n = \cos \frac{n\pi}{L} x, \quad n = 1, 2, \dots$$

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The corresponding T_n must satisfy the equation

$$T'_n(t) + \frac{n^2\pi^2}{L^2}kT_n(t) = 0.$$

For $n \geq 1$

$$T_n(t) = a_n e^{\frac{-n^2\pi^2}{L^2}kt}.$$

For $n = 0$

We have $T'_0(t) = 0$ and hence

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Superposing the product solutions, we obtain

$$u(x, t) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right) e^{\frac{-n^2\pi^2}{L^2}kt}$$

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The corresponding T_n must satisfy the equation

$$T_n'(t) + \frac{n^2\pi^2}{L^2}kT_n(t) = 0.$$

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Separation of variables

To finish the solution we must determine the coefficients a_n so that match the initial conditions. This yields

$$f(x) = a_0 + \sum_{n=1}^{\infty} a_n \cos\left(\frac{n\pi}{L}x\right)$$

Recognizing it as a cosine expansion of f we conclude that

$$a_0 = \frac{1}{L} \int_0^L f(x) dx \quad \text{and} \quad a_n = \frac{2}{L} \int_0^L f(x) \cos\left(\frac{n\pi}{L}x\right) dx$$

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Example

Example:

Consider an insulated wire of length 1 whose ends are also insulated. Let $k = 0.25$ and let the initial heat distribution be $u(x, 0) = 100x(1 - x)$. Find the temperature function $u(x, t)$.

Solution:

We are solving the following IBVP problem:

$$\begin{aligned}u_t &= 0.25 u_{xx}, \\u_x(0, t) &= u_x(1, t) = 0, \\u(x, 0) &= 100x(1 - x) \quad \text{for } 0 < x < 1.\end{aligned}$$

We have $k = \frac{1}{4}$, $L = 1$, and $f(x) = 100x(1 - x)$. Therefore

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The solution is then given by the series:

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